configuration has high entropy when densities are so low that we can ignore gravity, the situation is otherwise when gravity can't be ignored; then, such a uniform gas has extremely low entropy. In comparison with black holes, the diffuse, nearly uniform gas was in an extraordinarily low-entropy state. Ever since, in accordance with the second law of thermodynamics, the overall entropy of the universe has been gradually getting higher and higher; the overall, net amount of disorder has been gradually increasing. After about a billion years or so, gravity caused the primordial gas to clump, and the clumps ultimately formed stars, galaxies, and some lighter clumps that became planets. At least one such planet had a nearby star that provided a relatively low-entropy source of energy that allowed low-entropy life forms to evolve, and among such life forms there eventually was a chicken that laid an egg that found its way to your kitchen counter, and much to your chagrin that egg continued on the relentless trajectory to a higher entropic state by rolling off the counter and splattering on the floor. The egg splatters rather than unsplatters because it is carrying forward the drive toward higher entropy that was initiated by the extraordinarily low entropy state with which the universe began. Incredible order at the beginning is what started it all off, and we have been living through the gradual unfolding toward higher disorder ever since.

This is the stunning connection we've been leading up to for the entire chapter. *A splattering egg tells us something deep about the big bang.* It tells us that the big bang gave rise to an extraordinarily ordered nascent cosmos.

The same idea applies to all other examples. The reason why tossing the newly unbound pages of *War and Peace* into the air results in a state of higher entropy is that they *began* in such a highly ordered, low entropy form. Their initial ordered form made them ripe for entropy increase. By contrast, if the pages initially were totally out of numerical order, tossing them in the air would hardly make a difference, as far as entropy goes. So the question, once again, is: how did they become so ordered? Well, Tolstoy wrote them to be presented in that order and the printer and binder followed his instructions. And the highly ordered bodies and minds of Tolstoy and the book producers, which allowed them, in turn, to create a volume of such high order, can be explained by following the same chain of reasoning we just followed for an egg, once